COS344 – Computer Graphics

Homework Assignment Report

Miniature Golf Course: Dino Park Phuket

Team Member	Student Number
Ayush Beekum	u23596351
Shavir Vallabh	u23718146
Chinmayi Santhosh	u24585671
Jaitin Moodally	u23621372
Diya Narotam	u23533596

Initial Report Due: 30 April 2026

1. Background

1.1 About The Golf Course

The miniature golf course we have chosen for our project is Dino Park Mini Golf, located at Kata Beach in Phuket, Thailand. Specifically, the park is situated at Patak Soi 7, Karon, Muang, Phuket, adjacent to the Marina Phuket Resort, and acts as one of the most recognisable landmarks along the Kata Beach strip. The park covers approximately 6,400 m².

Dino Park Mini Golf opened in 1997 and was conceptualised and built in collaboration with the special effects team responsible for the dinosaur sequences in Jurassic Park (1993) and The Lost World: Jurassic Park (1997). This partnership made it the first dinosaur-themed attraction in Thailand, and quite possibly in all of Asia. The park has since grown to include animatronic life-sized dinosaur replicas, a central erupting volcano with light and sound effects, prehistoric caves, waterfalls, tropical jungle streams, and a Flintstones-inspired stone-age themed restaurant and bar. The 18-hole mini golf course winds through the entire prehistoric landscape, with each hole incorporating a unique dinosaur scene and terrain challenge.

Figure 1: An aerial map of the entire park

1.2 Why We Chose This Course

We chose Dino Park as the subject of our rendering for several reasons. Firstly, it complies with all the structural requirements of the assignment: it has exactly 18 holes, incorporates multiple height differences and inclines, features extensive aquatic elements (streams, waterfalls, and a prehistoric swamp), includes cave tunnels, and is adorned with numerous distinct decorations and prop types.

The courses we have played while conducting our research did not have a defining theme or unique layouts. This course on the other hand offers strong jungle theming with steep slopes, waterfalls, and even a volcano. The final feature in particular offers a unique lighting challenge, the lava providing a dynamic, colourful point light source that is sure to make our render stand out during nighttime.

Figures 3 & 4: Rigorous field research

This choice of course felt superior as it is something entirely unique, and comes with the added benefit of being designed to be used both during the day and night, which will allow us to create a striking render demonstrating a full range of OpenGL techniques.

2. Miniature Golf Course Layout

Dino Park occupies an irregularly shaped footprint of approximately 80 m × 80 m (using Google Maps satellite imagery and the park's published 4 rai land coverage as reference). The 18 holes are arranged along a winding path that flows through distinct biomes: a jungle entry zone (Holes 1–4), a swamp & waterfall zone (Holes 5–8), a cave & tunnel zone (Holes 9–12), a highland volcanic zone near the central volcano structure (Holes 13–16), and a final open clearing zone (Holes 17–18). The flow of the course follows a rough figure-8 to maximise use of the land while ensuring players always progress forward.

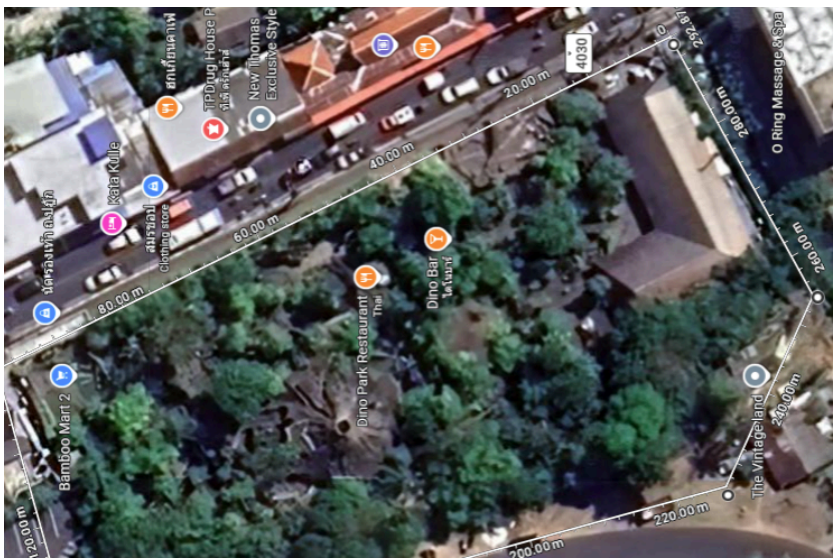


Figure 5: A satellite view of the golf course, 2026 Google Maps

2.1 Obstacles

The course incorporates several categories of obstacles across its 18 holes:

- **1. Dinosaur Statues:** Large-scale fibreglass and concrete replicas of Triceratops, Brontosaurus, Stegosaurus, Pterodactyl, and Tyrannosaurus Rex are placed adjacent to, behind, or straddling holes. These are quite complex to create, so we will not be adding every single dinosaur from the real-life course and only create one or max two.
- **2. Rock Formations:** Irregular volcanic basalt-style rocks are clustered around each hole boundary, serving both as aesthetic terrain and as physical obstacles that redirect the golf ball. We will incorporate our practical 4 code that relates to changing the surface of the golf ball to make the boundary walls made of rocks to be jagged using several circles along with rectangles to define the basic boundary structure of each hole.
- **3. Bone Obstacles:** Some holes feature decorative oversized dinosaur bone props placed across the fairway as direct ball obstacles. These will be rendered as elongated cylinders and will feature a chalky appearance.
- **4. Cave Tunnel** (Holes 9–12): A low-arch concrete cave structure covers several consecutive holes, requiring the ball to pass through the dark interior. The tunnel interior is rendered as a subdivided half-cylinder with a rock texture applied to the inner surface, to replicate the texture we will add rectangle blocks on the cylinder wall to give it more texture.
- **5. Central Volcano** (Holes 13–16): The centrepiece of the park is a large artificial volcano, which periodically erupts with sound effects and fog/flame simulation. In the render it is modelled as a truncated cone with a hollow caldera, and the eruption taking place we may implement as a possible bonus feature, but for now it will just be a normal static object in the centre of the course. The lava will serve as a point light source, illuminating its surroundings.
- **6. Waterfall Wall** (Holes 5–6): A tiered concrete waterfall cascades over rocks into a stream channel. In the render it is a sequence of thin quad strips with an animated water texture scrolling downward. For the bonus mark section we may implement the features to make the water look more realistic depending on the difficulty.
- **7. Swamp Hazard** (Hole 7): A shallow swamp pool surrounds part of Hole 7's fairway, acting as an out-of-bounds marker. Rendered as a flat plane with a murky-green semi-transparent texture, we will incorporate the transparency from practical 4 and possibly add light in the water for effect.
- **8. Bridge (Holes 6–7)**: A short wooden bridge crosses the stream between the waterfall zone and the swamp. Rendered as a flat platform with railing posts

(thin cylinders topped with small spheres) on either side and multiple rectangles for the floor to represent planks.

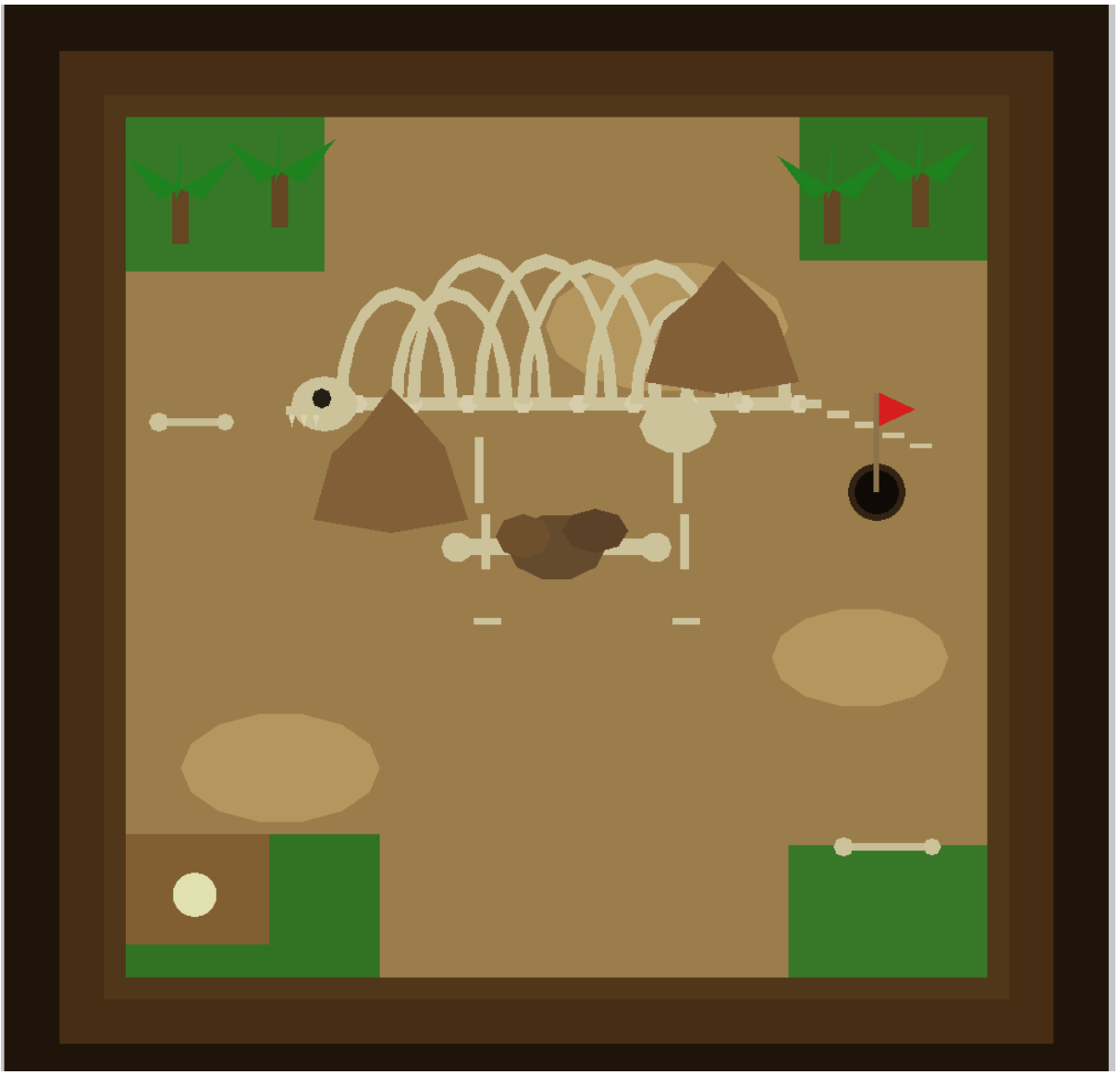


Figure 6: A top-down mockup of hole 6

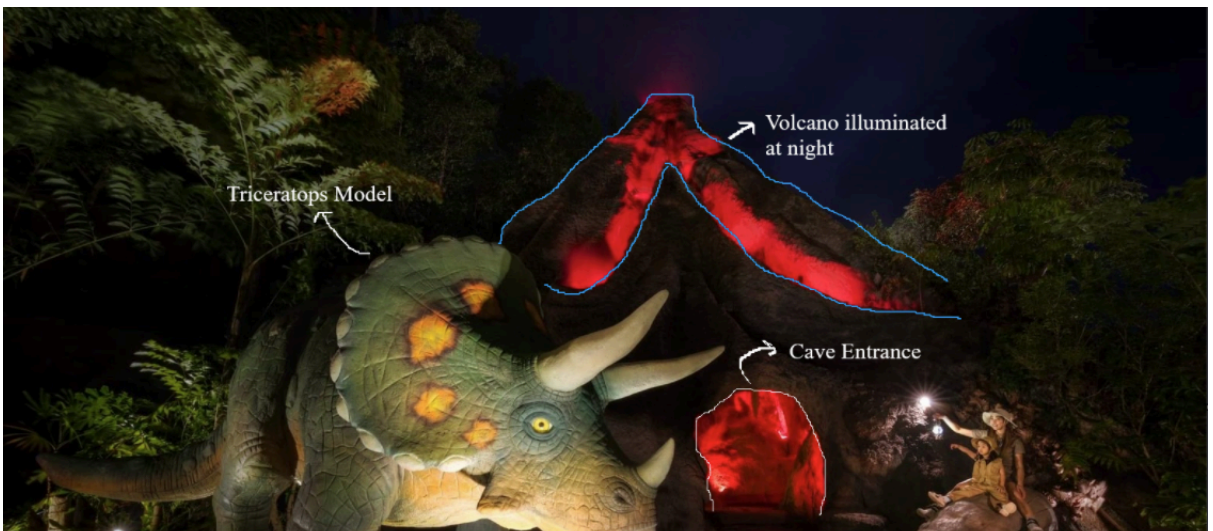


Figure 7: A view at night, showcasing some points of interest

Figure 8 : A daytime view from the other side of the volcano

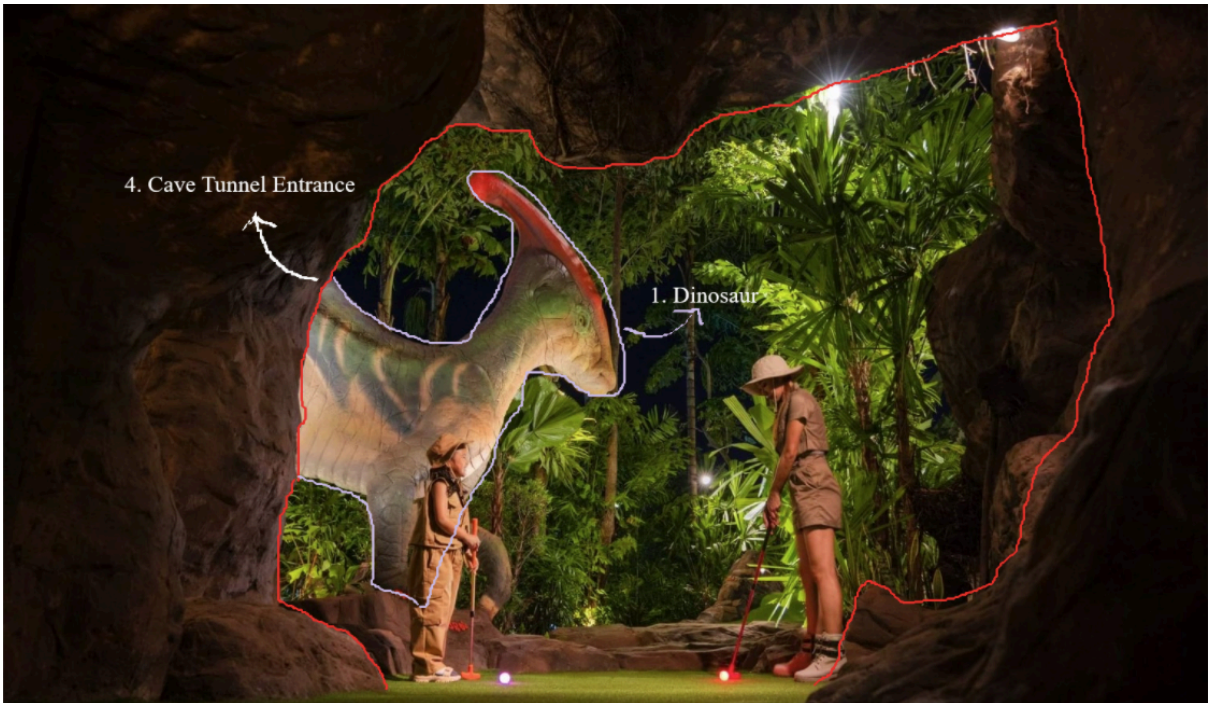


Figure 9: A view from inside the cave

2.2 Surfaces and Fairway Layouts

Each of the 18 holes is surfaced with artificial green turf carpeting in varying shades of bright green. We will use different shades of green for different parts of the course to showcase the different biomes in each section of the entire course. Boundary walls are created from casted concrete, and aim to mimic natural rock or logs. Modelling these organic surfaces will be challenging, keeping in mind a manageable poly count. These borders are tall enough to prevent balls from escaping. Some holes include deliberate 45° or 90° turns to reduce line-of-sight and increase difficulty.

Figure 10 : The fairway, featuring a slope and “natural’ geometry on the boundaries

2.3 Decorations and Vegetation

Surrounding the fairways are five or more distinct decoration categories:

- **Tropical Trees and Palms**: Palm trees, ferns, and tropical shrubs line the pathways between holes, creating the enclosed jungle atmosphere.
- **Directional Signage and Scorecards**: Wooden-style sign posts direct players to each hole. Rendered as a simple rectangle and triangle combined in brown.
- Pathway Lighting Poles: Short stone-effect columns with warm globe lights line the walking paths. These serve both as decoration and as light-emitting objects in the night-mode render and we will use ambient lighting for these globe lights in the render.
- **Flintstones Car Prop**: Near the park entrance stands the iconic Flintstones stone-wheel car prop It will be present in the render as a static decorative object, we will likely not be creating the park entrance itself but will incorporate it as part of the first hole.

3 Sky Box

3.1 Construction

The skybox will be implemented using a continuously rotating sphere, centered at the world origin, with a circumference of approximately five times the units as the actual course. This ensures that no visual clipping will occur, even if the drone ascends to its maximum altitude, and will create an immersive sense of distance. The depth buffer will be set to ignore depth writes for the skybox so it always appears in the far plane.

3.2 Texturing

We will need to use linear interpolation to determine the light level at each stage of the cycle. One large texture may be used, with the sun and moon positioned opposite from one another. Specific skybox shaders may be utilised around certain times, such as dawn/dusk to add a specific tint for a more immersive atmosphere. Toggling the time will simply rotate the sphere such that the appropriate celestial body, either the sun or moon, will appear directly above the render, and apply the appropriate shaders.

Figures 11 & 12 : An example skybox in Minecraft, featuring one continuous texture, with the sun opposite from the moon, and time-specific shaders for dusk/twilight.

4 Terrain:

4.1 Height Differences as Part of a Hole

Dino Park has multiple height transitions across its course to increase difficulty and create visual interest. The most prominent examples include:

- Uphill Ramp (Hole 3): A sloped section with a shallow incline rises over a boulder prop, requiring the player to account for gravitational deceleration so the ball does not roll back. Rendered as a quad plane with a light green carpet texture in the beginning.

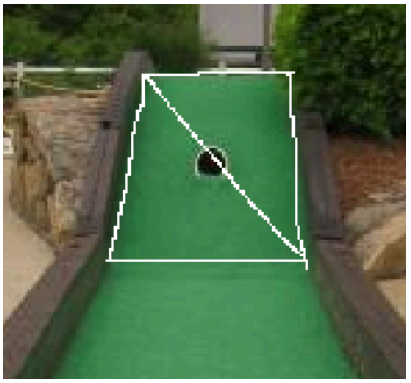


Figure 13 : An example of an uphill slope

- Downhill Drop (Hole 8): After the waterfall zone, the fairway descends via two stepped concrete platforms. The ball accelerates down the steps. Rendered as a staircase of three flat rectangular quads at descending Y-values.

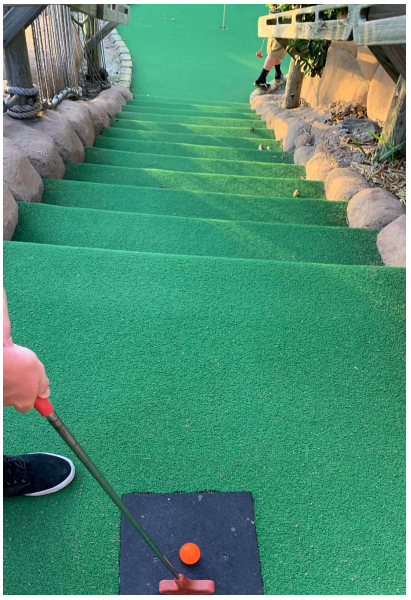


Figure 14 : Steps incorporated into the course

- Raised Island Green (Hole 14): The hole's cup is situated on a raised circular island platform reachable through a narrow inclined path. Rendered as a short

cylindrical pedestal topped by a flat circular fairway.

Figure 15 : An example of a raised hole

- Volcano Approach Ramp (Hole 15): The fairway rises along the base of the central volcano structure and curves around it. Rendered as a curved ramp section using a series of rotated and connected quad strips.

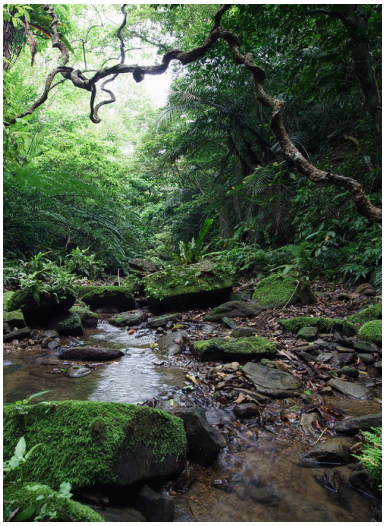
Figure 16 : A different course featuring a similar volcano

4.2 Terrain Differences as Course Aesthetic

Beyond the fairway surfaces themselves, several distinct terrain material zones exist across the course:

- **Artificial Turf (Fairways):** Bright green grass carpet, where as the further you go in the course the carpet will become a darker green. Rendered with a repeating grass texture.
- **Volcanic Rock Ground:** Between holes and around the volcano, dark grey-black basalt-textured ground covers the floor. Rendered with a rough rock normal map to simulate surface irregularity. Concrete Pathways: Smooth light-grey concrete paths link holes and serve as the visitor walkway. Rendered with a solid concrete texture.
- **Jungle Floor:** Under the tropical tree cover between holes, a dark organic ground texture is applied, incorporating leaf litter details.
- **Swamp Surface (Hole 7)**: Dark greenish-brown semi-transparent water plane.
- **Stream Bed (Holes 5–8):** Shallow running water over smooth river stones. Rendered as a semi-transparent water plane over a river-pebble texture on the substrate.
- **Lava Channels:** An emissive orange/red texture, with a glow effect. Will ‘flow’ throughout the volcano and outside it
- **Concrete Path:** Visitor walkways around the course, grey, chalky appearance. Will take on natural geometry to blend in to the environment

Figure 17 : An example of similar textures found in the golf course



Figures 18 - 22 : Real-life examples of the terrain we will model

5 Obstacles:

In addition to the terrain features discussed above, the following obstacle types are distributed across the 18 holes.

5.1 Dinosaur Statue Obstacles

Each hole features at least one dinosaur statue as both a decorative element and a physical barrier but due to the difficulty we will not implement one for every single hole. Two possible dinosaurs we may implement are as follows:

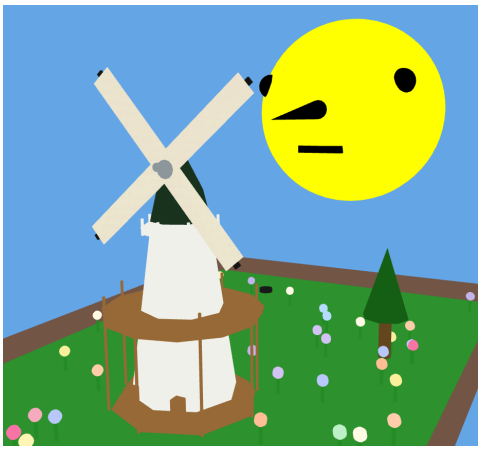
- Triceratops (Hole 1 – Entry): The three-horned frill and body block the left side of the fairway. Constructed from spheres (body, head), cylinders (legs, tail), cones (horns), and a flat circular frill disc.
- Tyrannosaurus Rex (Hole 16): The largest and most dramatic obstacle. Animatronic: head rotates on a servo. Rendered with a hinged head and periodic rotation animation in the render loop. Positioned beside Hole 16 with its jaw open over the cup area, for a bonus feature similar to making the windmill rotate in practical 3, we can use relatively similar code to make the head move.

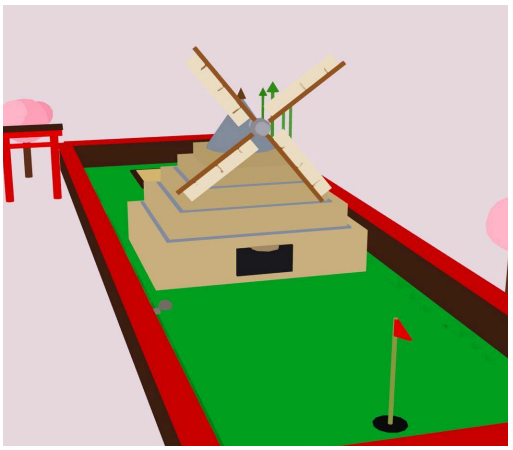
Figures 23 & 24 : Example dinosaur models

5.2 The Windmill

Although Dino Park does not feature a windmill in its real-world layout, the assignment spec requires one to be incorporated. We will position the windmill at Hole 4, in the transitional zone between the jungle entry and the swamp area. This location is chosen for two reasons: first, the hole at this point has a slightly wider fairway footprint, providing an ample amount of space for the windmill structure without obstructing gameplay significantly; second, the windmill's rotating blades will serve as a timing obstacle, requiring the player to putt through the gap in the rotating sails, a classic miniature golf challenge that heightens difficulty at this early stage of the course.

The windmill will be taken directly from the relevant team member's Practical 3 submission. It consists of a tapered rectangular tower body, four sail arms radiating from a central hub, and four rectangular sail panels. The sail rotation will be animated using a continuous rotation matrix applied around the forward-facing (Z) axis of the sail hub. The windmill will be thematically adapted with a stone-texture surface to blend with Dino Park's rocky





aesthetic.

Figure 25 & 26: Examples of teammate's windmills

6 Materials:

The rest of the course will be built from the standard green used on a mini golf course as our primary material for playable areas. Other materials used will include surfaces such as rock, water and the volcanic materials used to render the volcano.

6.1 Artificial Turf (Fairways)

Colour: Bright green. Diffuse component dominant; specular coefficient very low (near-matte surface). A repeating 2D grass blade texture is filled throughout the course. This material is used on all 18 fairway surfaces and the raised greens but the green colour will become darker throughout the course.

6.2 Volcanic Rock

Colour: Dark grey-black (RGB 50, 50, 50). Medium diffuse, very low specular, high roughness. A rock texture is applied to simulate surface irregularity. Used on rock formation obstacles, cave walls, and the volcano structure exterior.

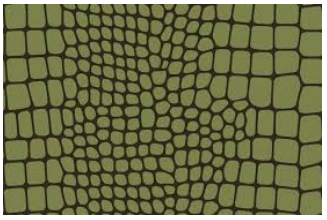
6.3 Concrete / Fairway Borders

Colour: Light grey (RGB 180, 180, 175). Low specular, smooth diffuse. A light concrete texture with a subtle is applied. Used on fairway border walls, cave floor, and pathway surfaces.

6.4 Water (Streams and Swamp)

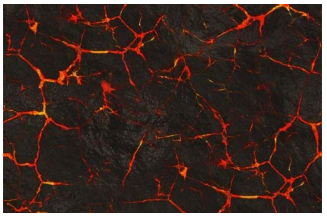
Colour: Semi-transparent blue-green (RGB 30, 100, 120, alpha 0.6 for stream; RGB 40, 80, 40, alpha 0.5 for swamp). Medium specular highlight to simulate surface reflectivity. Blended with the underlying substrate texture using alpha blending.

6.5 Dinosaur Statue Material



Colour: Varies per species — the T-Rex uses a dark olive-brown (RGB 80, 60, 40); the Triceratops uses a grey-green (RGB 90, 110, 80); Medium diffuse, low specular. Scaled-skin texture for a rough reptile hide surface.

6.6 Volcano Exterior

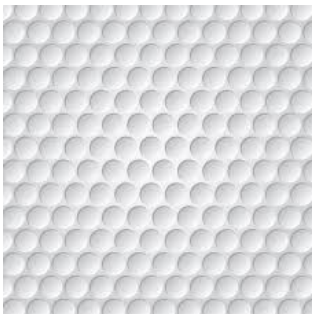


Colour: Dark charcoal exterior (RGB 40, 40, 40) with glowing orange-red lava streaks (RGB 220, 80, 10) around the caldera rim. The lava streaks are implemented as an emissive texture channel that ignores the standard lighting model, ensuring they remain visible in night mode. The overall shape uses a black rock texture with high-contrast orange emissive overlay.

6.7 Wood (Bridge, Signs, Windmill)

Colour: Medium brown (RGB 120, 70, 30). Medium diffuse, very low specular. A wooden rough texture with visible planks is applied. Used on the wooden bridge, directional signs, and the windmill tower.

6.8 Golf Ball



Colour: White (RGB 255, 255, 255). High specular component with a tight specular exponent to produce the characteristic bright highlight on a glossy ball surface. A subtle dimple normal map is applied to simulate the ball's surface textu`re. We can implement our practical 4 to give the ball texture and shape.

7. Lighting

Dino Park is an open-air course, so natural solar illumination dominates during the day. After dusk, the park's extensive decorative lighting transforms the atmosphere. The render will support two distinct lighting modes: Day Mode and Night Mode, toggled via a keyboard key (e.g., 'L').

7.1 Daytime Lighting

A single directional light source simulates the sun, positioned at a high angle consistent with a midday tropical sun. Its colour is a bright warm white (RGB 1.0, 0.97, 0.9). A global ambient term (RGB 0.3, 0.3, 0.35) approximates sky diffusion and fills shadows partially. Shadows will be rendered using shadow mapping from the directional light's perspective: the scene is first rendered into a depth framebuffer from the light's view, and the shadow map is then sampled in the main pass fragment shader to determine shadow coverage.

7.2 Nighttime Lighting

In night mode the directional sunlight is removed. The following point and spot lights are active:

- Pathway Globe Lights: Scattered warm point lights (colour RGB 1.0, 0.85, 0.6, intensity moderate) placed along the visitor walkways.t These produce soft overlapping pools of warm light on the ground plane and adjacent rocks.
- Volcano Glow: A strong point light at the caldera of the volcano (colour RGB 1.0, 0.45, 0.1) with high intensity and moderate attenuation, casting a warm orange-red glow on surrounding terrain and the dinosaur statues in holes 13–16.
- Hole Flag Lights: Each hole's flag pole has a small low-intensity cool-white point light (colour RGB 0.9, 0.95, 1.0) illuminating the cup area to ensure visibility during play.
- Ambient Night Term: Reduced global ambient (RGB 0.05, 0.05, 0.1) simulating faint moonlight and light pollution from the nearby Kata beach.

7.3 Shadows

Shadow mapping will be implemented for the primary directional light (daytime) and the volcano point light (nighttime). The shadow pass renders the scene geometry into a depth texture from the light's viewpoint. In the main render pass, each fragment's position is transformed into the light's clip space and compared against the shadow map depth value. Fragments in shadow receive only ambient illumination, while lit fragments receive the full contribution.

8. Drone

The drone serves as the user's view point and primary way of navigating through the rendered golf course. It is modelled as a quadcopter with a camera gimbal mounted on the underside.

8.1 Geometry

The drone body is a flattened octagonal prism (rendered from flat-shaded polygon faces). Four arms extend diagonally from the body, each terminating in a motor housing (short cylinder) topped by a propeller (two thin rectangular quads per propeller, angled slightly). The camera gimbal is a small sphere beneath the body with a torus ring representing the stabilisation frame. The drone will be rendered in third-person view when the user presses the 'V' key, revealing the drone model; otherwise the camera is in first-person (drone camera perspective) mode. We can implement the windmill sails from practical 3 for the blades of the drone to show it spins to portray it flying.

8.2 Controls

The drone is controlled via keyboard and mouse:

- W / S: Pitch forward / backward (tilt and translate in the drone's local forward direction).
- A / D: Yaw left / right (rotate around world-up Y-axis).
- Q / E: Roll left / right (tilt along local forward axis, affecting lateral translation).
- Space / Left Shift: Ascend / descend along world-up axis.
- Mouse movement: Adjust the camera pitch and yaw (independent of drone body orientation in first-person mode).
- V: Toggle between first-person (FPV) and third-person drone view.
- L: Toggle between Day and Night lighting modes.
- Esc: Exit the application.

8.3 Projection

A perspective projection matrix is used at all times, with a field of view of 60°, an aspect ratio matching the window dimensions, a near plane at 0.1 units, and a far plane at 1000 units. The drone's view matrix is computed from its current position and orientation using a lookAt construction. Zooming is implemented by reducing the FoV parameter when the user holds the 'Z' key (optional feature for bonus marks possibly), transitioning from 60° to 20°.

8.4 Additional Features

The drone will have two small red LED point lights (one on each front motor arm) and two green LED point lights (one on each rear motor arm), implemented as low-intensity point lights that cast a subtle glow on nearby surfaces in night mode. In third-person view, the drone will cast a shadow on the terrain below too.

9 Code Design:

9.1 Architectural Overview

The codebase is organised as a layered system with five main modules; *core, scene, lighting, objects* and *camera*. Each of these maps directly to a directory in our repository, allowing for clear responsibilities and separation of concerns. The *main.cpp* entry point will initialise the GLFW window and the OpenGL context, and then hand over objects to *scene* which owns and orchestrates each renderable entity, some of which may be blender models located under 'assets'.

We aim to keep our main relatively low on code, thus allowing us to test and implement in isolation and maintain a more consistent, readable codebase.

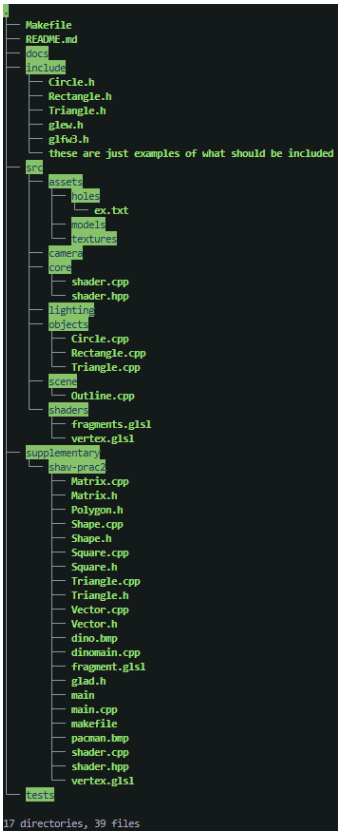


Figure 27 : tree run against the root of our skeleton repo

9.2 Data Structures Used

std::vector will be the primary container used throughout, serving the same purpose as in the majority of our practical assignments. Every collection of variable length objects, including abstractions (groups of other objects), such as each GolfHole, Obstacle objects, LightSource objects. This is the obvious choice for this use case, with random access in O(1) time, cache friendly iteration (since it uses contiguous memory), and size only ever being modified during startup for a static scene. This gives us inherent access to its iterator as well, which is useful for ordered rendering.

A linked list was considered for this use case, however we do not need insertion or deletion in the middle of a frame, so its benefits are not realised here despite the additional hassle in using one.

std::map<std::string, GLuint> may be used for texture caching. We can maintain a map from the file path to a texture handle. This works almost exactly like the *Flyweight* pattern; when a texture already exists and an object requires it, it is fetched by using the map. Otherwise, a new entry in the map is made, and the texture is cached.

9.3 Design Patterns Used

Template method may be used to create consistent objects that may all be drawn and rendered the exact same way. A generalisation class, *Object*, may declare pure virtual functions to update and draw. Every concrete class, for instance Terrain, Volcano, Drone, etc. will implement these functions, allowing our SceneBuilder to use one interface to create all of these objects.

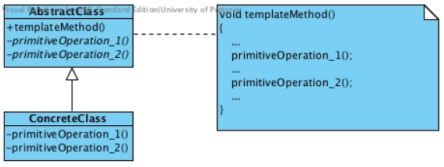


Figure 28 : Template Method

An implicit tree of scene objects was considered for large compositions of objects, but we realised that the *Composite* pattern would be much better suited for this purpose.

We could have complex scene objects such as GolfHole, that themselves comprise of many smaller objects, such as ramps, fairway sections, boulders, etc. This abstraction allows us to treat these large objects and smaller individuals the exact same, simplifying function calls in our renderer to only call for an update or draw on the top level objects.

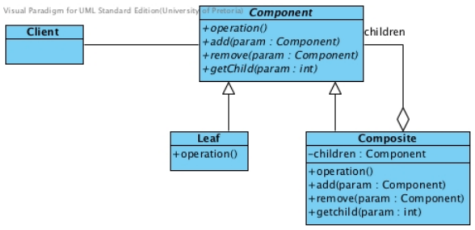


Figure 29 : Composite

In terms of creating these complex structures, the *Builder* pattern would enable this behaviour cleanly. Instead of bloated constructors, we can employ this pattern to create the object cleanly. This has the added benefit of allowing us to easily change the order in which objects are added to the scene, preventing confusion such as water rendering before the ground.

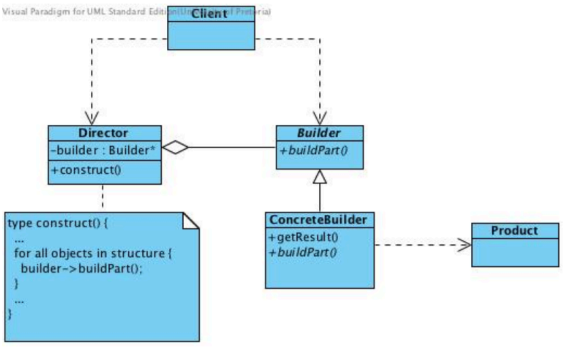


Figure 30 : Builder

10. Bonus Marks

10.1 Animated Water [5 marks]

The stream, waterfall, and swamp water surfaces will feature animated movement. A time-varying UV offset is applied each frame to a scrolling normal map on all water planes, producing the appearance of flowing current. The waterfall uses a fast downward scroll; the

stream uses a slower scroll; the swamp uses a slow circular swirl. This gives each water body a visually distinct motion to differentiate between the different bodies of water.

Figure 31 : Water, the defining feature of Skyrim Special Edition

10.2 Day/Night Cycle Progression [5 marks]

Rather than an instant toggle, a constant day/night cycle will be implemented. The sun's directional light direction and colour smoothly interpolate over a configurable cycle period (default 5 minutes real time, may change to be a shorter time for demo purposes). The skybox texture is cross-faded between the daytime and nighttime sets using a blend factor uniform in the skybox shader. Pathway lights and volcano glow gradually increase in intensity as the sun descends, and decrease as it rises. This produces a continuous, natural-feeling progression of time across the course. (a toggle will still be implemented to showcase day and night without having to wait).

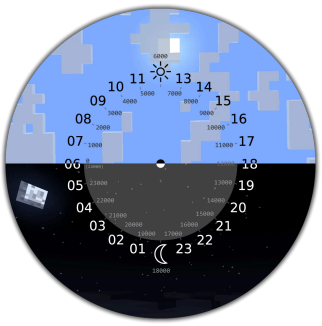


Figure 32: A representation of the day/night cycle

10.3 Flight Information HUD [5 marks]

A HUD will be rendered in screen-space showing the drone's current altitude (Y position), forward speed (magnitude of velocity vector), and orientation (Euler angles or compass heading). The HUD will be toggleable via the 'H' key.

Figure 33: An example HUD

10.4 Colour Filters on Drone Camera [5 marks]

A post-processing step will be implemented using a framebuffer object (FBO). The scene is first rendered to an off-screen colour texture. A fullscreen quad then applies a fragment shader that transforms the colour using one of several filter modes: Greyscale (dot product with luminance weights), Inverted (1.0 - colour), Sepia (matrix transform), and Night Vision (green-tinted greyscale with screen-space noise). The active filter is cycled through using the 'F' key.

10.5 Minimap [5 marks]

A top-down minimap of the course will be displayed in the bottom-right corner of the screen. It is rendered by placing a secondary orthographic camera above the course (looking straight down) and rendering the scene geometry to a small viewport. A marker icon (small coloured triangle) indicates the drone's current position on the minimap. The minimap is composited onto the main view using a second FBO and a screen-space quad. It can be toggled through the 'M' key.

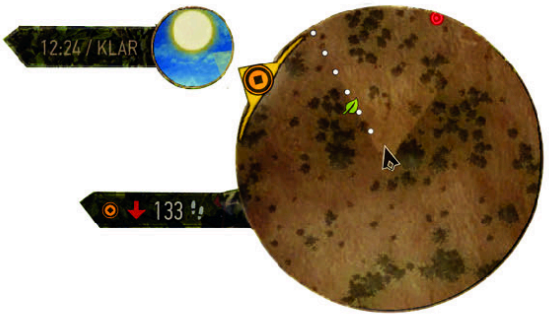


Figure 34: An example minimap

10.6 5 Additional Complex Objects [5 marks]

Five extra complex decorative objects will be added to enrich the course beyond the baseline requirements: (1) an animatronic T-Rex in a standing pose near Hole 1 or 2; (2) a detailed tropical palm tree with a procedurally-placed frond arrangement; (3) a stone tiki torch prop; (4) a Flintstones-style car at the beginning of the course; and (5) the huge volcano in the centre of the course simulating a cool jurassic feel.

Figure 35: The Flinstones car we will attempt to model

10.7 Collision Detection

Collision detection will be implemented to allow realistic interaction between the ball and environment. We will use simplified bounding boxes to ensure efficiency. When a collision is detected, the ball's velocity will be adjusted using basic reflection principles, behaving similar to the basic ray tracing that we covered towards the start of the year.

10.8 Advanced Lighting

Lighting in this course is a main focal point, from the volcano centerpiece to the constant day/night cycle. We may incorporate Blinn-Phong shading to calculate ambient, specular and diffuse components. These enhancements will greatly augment the atmosphere of the course, as the real life counterpart focuses on this aspect heavily.

10.9 Loading From Text Files

To improve flexibility and ease the development process, we will implement a system that loads hole and object layouts from structured text files, where each file specifies the object's types, positions, scale, and orientations in a predefined format. The program will parse these during runtime, decoupling scene composition from logic, reducing the need for recompilation for design changes.

10.10 Playable Render

After implementing collision detection and smooth movement, the render may be expanded into a fully playable experience by allowing user interaction with the ball, and possibly traversal of the course.